RAJALAKSHMI ENGINEERING COLLEGE

B.E. COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

CR23721 - PROJECT ABSTRACT (2026-27)

TITLE	Prompt Shield: AI based prompt injection and recovery system
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ABSTRACT:

Large Language Models (LLMs) such as ChatGPT, Gemini, and Llama have become widely used in various applications, but they are vulnerable to prompt injection attacks, where malicious prompts manipulate the model to ignore system instructions, reveal sensitive information, or generate unintended responses. Existing solutions primarily focus on detecting and blocking these attacks, which can also prevent legitimate user requests from being processed. To address this limitation, this project proposes Prompt Shield: AI-Based Prompt Injection and Recovery System, a secure framework that not only detects prompt injection attacks but also recovers safe user prompts.

The system uses DeBERTa-v3 to detect malicious prompts, Regular Expressions (Regex) to sanitize harmful instructions, FLAN-T5 to repair the prompt while preserving its original meaning, and Sentence-BERT (SBERT) to verify semantic similarity between the original and repaired prompts. A rule-based risk scoring mechanism further classifies the severity of detected attacks before forwarding the sanitized prompt to the LLM. By combining detection, sanitization, repair, and intent verification, the proposed system enhances the security, reliability, and usability of LLM-based applications while ensuring that genuine user intent is preserved.

PROJECT GUIDE **PROJECT COORDINATOR**